INTERNSHIP TASK REPORT: TASK-1

1. TASK INFORMATION

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Task Number: Task-1: Foundations of Cybersecurity

Submission Date: 13/06/2026

2. TASK OBJECTIVE

The primary objective of this task is to build a solid foundation in cybersecurity principles, networking, and cryptography, while successfully configuring a functional professional hacking laboratory environment.

3. LAB ENVIRONMENT CONFIGURATION

- **Virtualization Platform:** Installed and configured VirtualBox/VMware to host the lab environment.
- **Attacker Machine:** Deployed Kali Linux to serve as the primary penetration testing environment.
- **Target Machine:** Deployed Metasploitable2/DVWA (Damn Vulnerable Web App) to act as target systems for testing.
- **Network Setup:** Configured a secure private lab network using a Host-Only Adapter to isolate traffic.

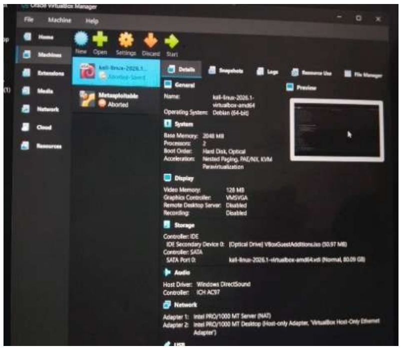
4. CORE COMPETENCIES COVERED

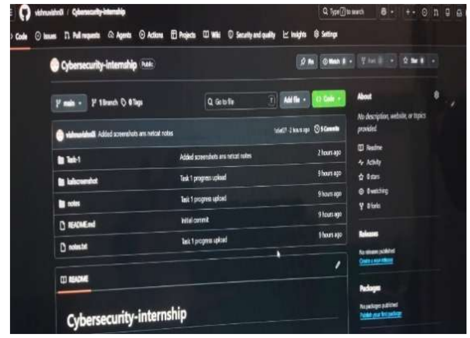
- **Cybersecurity Fundamentals:** Analyzed the CIA Triad (Confidentiality, Integrity, Availability) and researched common threat types including Phishing, Malware, DDoS, SQL Injection, Brute Force, and Ransomware.
- **Attack Vectors:** Examined methodologies behind Social Engineering, Wireless Attacks, and Insider Threats.
- **Linux System Administration:** Practiced file system navigation, directory permissions (chmod/chown), package management, and essential networking commands such as ifconfig, ping, netstat, and traceroute.
- **Network Fundamentals:** Studied the OSI Model, TCP/IP protocol suite, DNS, HTTP/HTTPS, IP addressing, Subnetting, and NAT.
- **Cryptography:** Explored the differences between symmetric and asymmetric encryption, hashing algorithms (MD5, SHA256), and digital certificates. Conducted hands-on exercises encrypting and decrypting messages using OpenSSL.

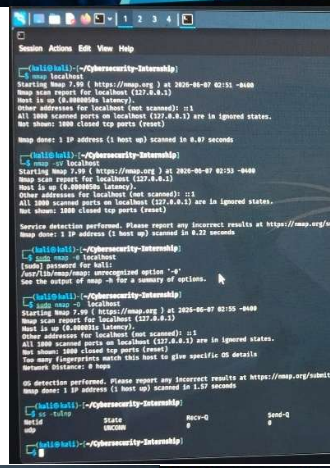
- **Security Tooling:** Gained proficiency in using Wireshark for packet capture, Nmap for network scanning, Burp Suite for web proxying, and Netcat for network debugging.

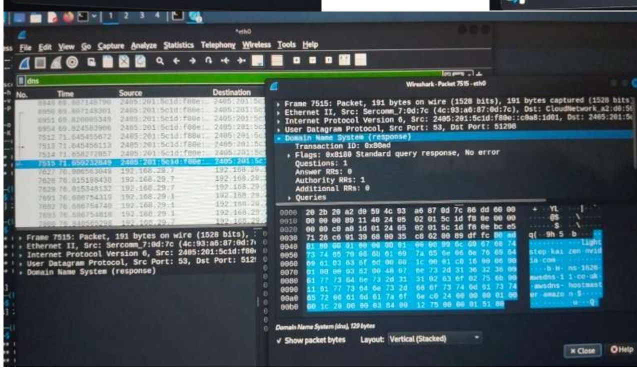
5. SUBMISSION DELIVERABLES

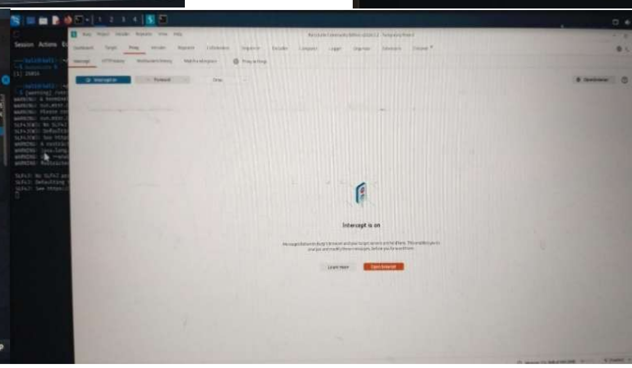
- **Lab Setup Report:** Documentation including screenshots verifying the successful installation and configuration of Kali Linux, the target machine, and a successful test packet capture via Wireshark.











- **GitHub Repository:** Public repository created containing comprehensive notes, the Linux cheat-sheet, and all relevant configuration documentation.
- vishnuvishn0i/Cybersecurity-internship
- **Video Walkthrough:** A 5-minute video demonstration recorded and uploaded to the "Featured" section of LinkedIn.
- https://www.linkedin.com/posts/vishnubishn0i_successfully-completed-task-1-of-cybersecurity-ugcPost-7471630948460310528-Hk-S/?utm_source=social_share_send&utm_medium=member_desktop_web&rcm=ACoAAGegcQsBI2gfxNpOlgcOtegGJxYFhELyPl8